Data Science 500 Assignment

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Name and Surname: _____

Student ITS No: _____

Qualification: _____ **Year of Study:** _____ **Semester:** _____

Assignment due date: _____ **Date submitted:** _____

QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: _____ **Date:** _____

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Book Market Analytics – Books to Scrape

(100 marks)

Project Description

You have been hired as a Junior Data Scientist to assist a publishing analytics team that specialises in helping bookstores and online retailers optimise their catalogue offerings. The team wants to understand the factors that influence book prices and ratings on Books to Scrape (books.toscrape.com), a sandbox website designed specifically for web scraping practice.

Your task is to build a complete end-to-end data science solution that can help the team analyse book catalogue trends and predict pricing patterns across different genres and rating levels.You will independently gather real-world structured data, process it into a usable form, explore catalogue behaviour, and build a predictive model that estimates book price based on publicly available book information.

This project simulates the workflow expected in real data industry roles.

Project Requirements

As the project owner, you must design and execute a workflow that includes:

1. ***Data Acquisition***

You will obtain your dataset directly from the Books to Scrape website (https://books.toscrape.com) by scraping all available product listings across the entire catalogue or within a chosen genre category (e.g., Mystery, Science Fiction, Romance, History).
You must use BeautifulSoup to scrape the data and extract at least the following information for each product:

- Book title
- Price (in GBP)
- Star rating (1–5, expressed as a word e.g. “One” to “Five”)
- Stock availability (In stock / Out of stock and quantity where shown)

You must collect at least 200 books (the full catalogue contains 1 000 books across 50 genres, so scraping multiple pages or the entire site is encouraged).

2. ***Data Cleaning***

You will convert raw, messy, real-world product listings into a clean, structured analytical dataset suitable for modelling. This includes handling missing or inconsistent entries, and preparing text fields (e.g., descriptions and tags) for analysis.

3. ***Data Visualization***

You are expected to investigate the behaviour of books within your chosen genre or across the full catalogue by creating visual representations of price patterns, relationships between variables, and emerging trends. Your visualisations should help reveal insights that retailers and publishers can use to understand what factors may influence book pricing and popularity.

4. ***Predictive Modelling***

You will build a linear regression model capable of estimating book prices from the features you derived from the Books to Scrape data.

5. ***Professional Reporting***

You will communicate your full project work in a concise, analytical report suitable for a consulting audience, highlighting:

- key insights about the book catalogue and pricing landscape,
- modelling outcomes,
- the practical implications of your findings, and
- recommendations for retailers, publishers, and data-driven book buyers.

Project Deliverables

You must submit:

1. **A complete Jupyter Notebook** documenting the entire project pipeline from data acquisition to modelling and interpretation.

2. **Your final dataset** constructed from your scraping process.
3. **A professional project report** presenting your methodology, insights, and conclusions.

Marking Rubric

1. Data Acquisition (Web Scraping) - 20 marks

Criteria	Marks
Correct use of BeautifulSoup to scrape Books to Scrape data	5
Extraction of all required fields	7
Adequate dataset size (minimum ~100 products)	5
Code clarity, structure, and reproducibility	3

Subtotal: 20 marks

2. Data Cleaning & Preparation - 20 marks

Criteria	Marks
Clean, structured dataset produced from raw scraped data	7
Appropriate handling of missing/inconsistent values	5
Proper formatting of numeric and text fields	4
Logical and well-justified cleaning decisions	4

Subtotal: 20 marks

3. Data Visualization & Exploration - 20 marks

Criteria	Marks
Relevant visualisations exploring book pricing, ratings, and genre trends	8
Exploration of relationships between key variables	5
Readability, correctness, and appropriateness of plots	4
Insightful interpretation of visual findings	3

Subtotal: 20 marks

4. Predictive Modelling (Linear Regression) - 20 marks

Criteria	Marks
Correct construction of modelling dataset	4
Accurate implementation of Linear Regression	4
Appropriate choice and justification of features	4
Meaningful evaluation using suitable metrics	4
Interpretation of results and model limitations	4

Subtotal: 20 marks

5. Report Quality & Communication - 20 marks

Criteria	Marks
Clear explanation of methods and workflow	8
Logical structure, clarity, and professional writing	6
Practical insights and recommendations	6

Subtotal: 20 marks